

- TP 3 : Load Balancer with Azure
- Professor: Ouassim Karrakchou
- Level : 4th year

TP 3

Load Balancer with Azure

Objectives:

The goal of this TP is to introduce you to a website deployment in Azure based on load-balanced VMs.

We will do the following tasks:

- Create a virtual network in Azure.
- Create 2 VMs in this virtual network.
- Add a bastion to allow secure SSH access to your VMs.
- Add a NAT gateway to allow the VMs to connect to the Internet.
- Install the Apache 2 web server on these VMs.
- Add a load balancer to handle the HTTP traffic and distribute it to your VMs.

Warning: some services we will use cost some credit money. Make sure you delete your resource group at the end of the lab!

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

- TP 3 : Load Balancer with Azure
- Professor: Ouassim Karrakchou
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0 - Check your allowed Azure region:

Before creating the resources of this Lab, you first have to identify a region where the required VM size is available.

1. Go to the Azure Portal and open **Policy** → **Assignments**:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade.MenuView/~/_Assignments

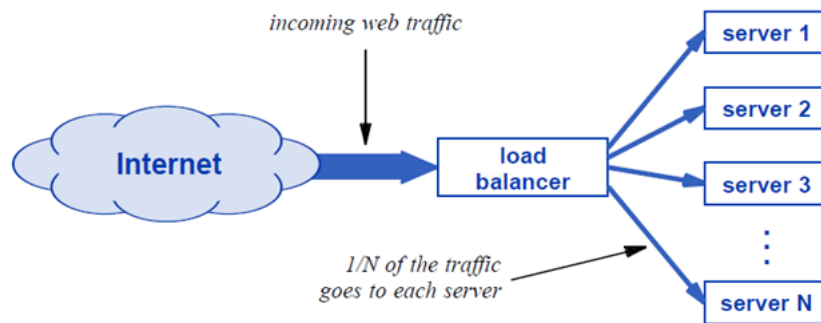
2. Check the **Allowed locations** policy and note the regions available for your subscription.
3. Go to **Virtual Machines** and start creating a new VM.
4. For each allowed region, use the following configuration:
 - **Region:** select one of your allowed regions.
 - **Security type:** Standard.
 - **Availability options:** No infrastructure redundancy required.
 - **Size:** click on **See all sizes** and search for Standard_B2ats_v2.
5. If B2ats_v2 is not available, try another one of your allowed regions.
6. When you find a region where B2ats_v2 is available, keep this region for the rest of the Lab.
7. Do not create the VM yet.

Important: The Virtual Network and the VMs must be created in the same region. If the Virtual Network is created in a different region, it will not appear when configuring the VM.

- TP 3 : Load Balancer with Azure
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I- Create a resource group:

In this TP, we want to create the load balanced VMs architecture we saw in the course to deploy a manually scalable website:



We will be using several Azure services. As usual, when using an Azure service, you have first to create a resource group that will regroup all the resources we create for a specific application:

1. Go to the Azure portal and create a resource group called **tp_load_balancer**. For the Region, select the Lab Region that you identified in Section 0.
2. Call the professor to show him when the resource group is created.

II- Create a Virtual Network:

Our first step is to create a virtual network that will contain all the VMs that will be used by the load balancer:

1. Go to the Azure Portal and search for **Virtual Network**:



Virtual network

2. Create a new virtual network with the following settings:
 - Resource Group: **tp_load_balancer**
 - Name: **vmNetwork**
 - Region: select the **Lab Region** that you identified in Section 0.
 - Security: leave everything unchecked

- TP 3 : Load Balancer with Azure
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- IP addresses:
 1. Go to Address space and delete the address space created by default.
 2. Create a new address space: **10.0.0.0/16**.
 3. Click on Add a subnet.
 4. Use the following configuration:
 - Subnet name: **vmSubnet**
 - IPv4 address range: select the **10.0.0.0/16** address space you just created.
 - Subnet address range: **10.0.0.0/24**
 - Network security group: create a new network security group called **vmNSG**.
 - Leave everything else by default.
- Il the professor to show him **before clicking on Add and Create**.

III- Create a first VM:

You will now create a first VM called VM1 that you will put in the virtual network you just created. This VM won't have any public IP address, so it won't be accessible from the internet, only from inside the virtual network.

1. Create a new Azure VM with the following settings:
 - Resource Group: **tp_load_balancer**
 - Name: **VM1**
 - Region: select the **Lab Region** that you identified in Section 0.
 - Availability options: **No infrastructure redundancy required**
 - Security type: **Standard**
 - Image: **Ubuntu Server 24.04 LTS - x64 Gen2**
 - VM architecture: **x64**
 - Size: **Standard_B2ats_v2**
 - Authentication: **SSH public key**
 - Public inbound ports: **None**

- TP 3 : Load Balancer with Azure
- Professor: Ouassim Karrakchou
- Level : 4th year

- Hard Drive (Disques): leave everything by default
 - Networking:
 1. Virtual Network: choose **vmNetwork**
 2. Subnet: choose **vmSubnet**
 3. Public IP: **None**
 4. NIC network security group: select **Advanced Parameters**, then select the network security group **vmNSG** you created in the last part.
 - Leave everything else by default.
2. Call the professor to show him **before clicking on Create**.
 3. After you click on create, **make sure you download the SSH private key** of your VM, you'll need it to connect to the VM.

IV- Create a Bastion:

The VM we just created is not accessible from the Internet. To connect to it in order to install the web server on it, we need to create a special host in a DMZ subnet of the virtual network that will have SSH access to the VMs. This special host will be exposed to the Internet and will be used as an intermediary to connect to our VMs. The goal is to avoid exposing critical servers to the Internet. This special host is called a **Bastion** host.

1. Go to the **vmNetwork** resource page in the Azure Portal.
2. Click on **Subnets** (sous-réseaux) in the left menu, then click on **+ subnet** to create the new **DMZ subnet** with the following configuration:
 - Subnet purpose: **Azure Bastion**
 - Starting address: **10.0.1.0**
 - Size: **/24**
 - Leave everything else by default.
 - Click on **Add**.
3. In the same left menu, click on **Bastion**, then click on **Deploy Bastion**.
4. Call the professor to show him.

The deployment of the Bastion takes around 10 minutes. In the meantime, go to the next part.

V- Create the second VM:

1. Using a similar procedure to what you did in part III, create a new VM called **VM2** with the same configuration.
2. Call the professor to show him **before clicking on Create**.

VI- Connect to the VMs using the bastion:

Wait for the bastion to finish deploying, then you'll be able to SSH to your VMs:

1. Go to the **VM1** page in the Azure portal.
2. Click on **Bastion** in the left menu. For **Authentication Type**, select **Private Key from Local File**, then enter your login (**azureuser** by default) and browse to the private key file of **VM1** you downloaded when you created it.
3. A new popup window should open with a terminal on VM1.
4. Do the same for **VM2** to open an SSH terminal on it through the bastion.
5. Call the professor to show him.

VII- Add a NAT gateway:

The VMs you have created are isolated in a virtual network. We will need to add a NAT gateway to their subnet to explicitly give them an outbound connection to the Internet that will be used to install the Apache 2 web server:

1. Go to the Azure Portal and search for **NAT Gateways**:



2. Create a new NAT gateway with the following configuration:
 - Resource Group: **tp_load_balancer**
 - Name: **vmNAT**
 - Outbound IP Addresses:
 1. Create a new public IP address called **natIP**

- TP 3 : Load Balancer with Azure
- Professor: Ouassim Karrakchou
- Level : 4th year

- Virtual Network: choose **vmNetwork**
 - Leave everything else by default.
3. Call the professor to show him **before clicking on Create**.
 4. You now have to add the NAT to the subnet of the VMs:
 - Got to the **vmNetwork** page in Azure Portal
 - Click on **Subnets** in the left menu, and click on **vmSubnet**
 - Select your **vmNAT** gateway and save.
 5. Got to the SSH terminal of VM1 and VM2, and execute the following command: `curl ifconfig.me`
 6. The command should return the public IP address of your NAT gateway.
 7. Call the professor to show him.

VIII- Install the Apache 2 web server in the VMs:

1. Go to the SSH terminal of VM1.
2. Execute the following command to install Apache: `sudo apt install apache2`
3. After it finishes, run the following command to overwrite the default web page of Apache with a page that tells you which VM served your request:
`echo "Hello from $HOSTNAME" | sudo tee /var/www/html/index.html`
4. Check that it worked by querying the web server using the following command: `curl localhost`
5. Redo the same steps to install the web server on VM2.
6. Call the professor to show him.

VIII- Create a load balancer to distribute the HTTP requests:

The two VMs are now correctly configured, we just need to create a load balancer to expose them to the Internet:

1. Go to the Azure Portal and search for **Load Balancers**:



2. Create a new Standard load balancer with the following configuration:
 - Resource Group: **tp_load_balancer**
 - Name: **vmLoadBalancer**

- TP 3 : Load Balancer with Azure
- Professor: Ouassim Karrakchou
- Level : 4th year

- Reference (SKU): **Standard**
 - Type: **Public**
 - Level: **Regional**
 - Front-end IP: create a new public IP for the load balancer called **loadBalancerIP**
 - Backend Pool: create a new backend pool called **vmBackendPool**. Select the **vmNetwork** virtual network and add your VMs under **resources**.
 - Load Balancing Rule (règle d'équilibrage de charge): create a new load balancing rule called **HTTPBalancing** that will distribute HTTP request received on port 80 (HTTP port) of the load balancer public IP to the VMs in the backend pool. Use the following configuration:
 1. Front-end IP: loadBalancerIP
 2. Main Pool (pool principal): vmBackendPool
 3. Protocol: TCP
 4. Port and Backend Port : 80
 5. Health Probe (Sonde d'intégrité): Create a new health probe called **httpHealthProbe** on **TCP port 80** with an **interval of 5 seconds**. The goal of this health probe is to check every 5 seconds if the web servers in the VMs of the backend pool are up and running.
 - Leave everything else by default.
3. Call the professor to show him **before clicking on Create**.
 4. Go to the page of the **vmLoadBalancer** resource you just created to retrieve its public IP address.
 5. Put the load balancer IP address in the web browser. It doesn't work. Do you know why?

IX- Allow HTTP traffic:

The issue comes from the network security group firewall we created at the beginning of the TP. It is currently configured to block everything. We have to add in it a rule to allow HTTP traffic.

1. Go to the page of the **vmNSG** network security group in the Azure Portal.

- TP 3 : Load Balancer with Azure
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2. Click on Inbound Traffic Security Rules (règles de sécurité de trafic entrant) in the left menu.
3. Add a new rule to **allow HTTP traffic**.
4. Try again to access the IP address of the load balancer in the web browser. Does it work now?
5. Call the professor to show him.

X- Delete your resource group:

Now that everything works, go to the Azure portal and delete the **tp_load_balancer** resource group.

XI- Assignment:

Based on what you learned in the TP explain the different steps needed to create a manually scalable website using load-balanced VMs. **Make sure you explain the purpose of every Azure service you used.** Use screenshots from the Azure portal. When you have finished, make sure you delete all the resources you used (show the steps).

How would you scale up your website to allow it to handle more requests?